Computer Networks: Assignment-8

Name: Suraj Shah

Admission Number: I24AI012

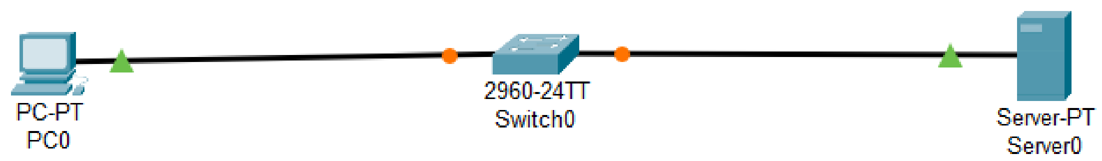
Q 1) Configure and implement a network using Packet Tracer and configure DNS.

a. Connect a PC with the server.

b. Sending Simple Text Messages in Realtime Mode

c. Establishing a Web Server Connection Using the PC's Web Browser

Diagram:



IP Config:

PC0: 192.168.1.2 / 255.255.255.0 / 192.168.1.3

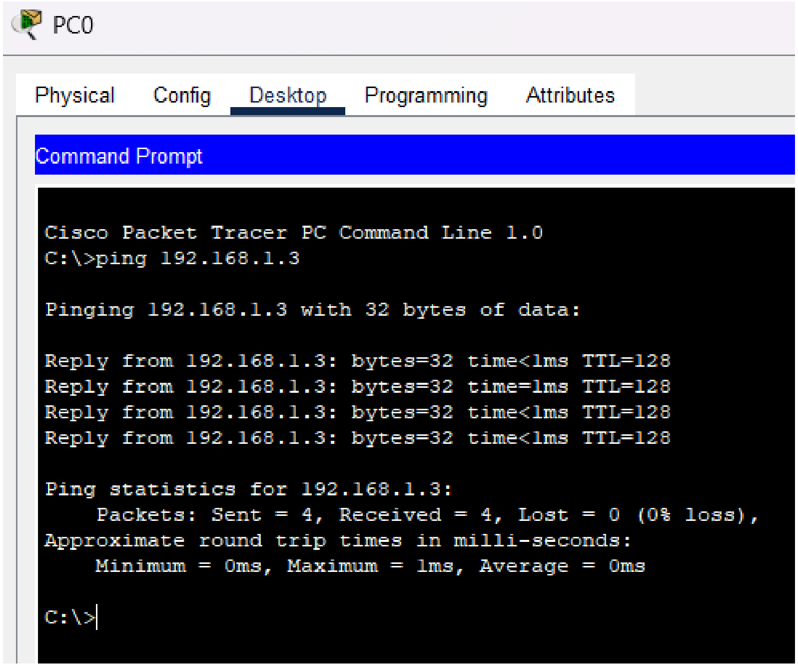
Server0: 192.168.1.3 / 255.255.255.0

DNS Config:

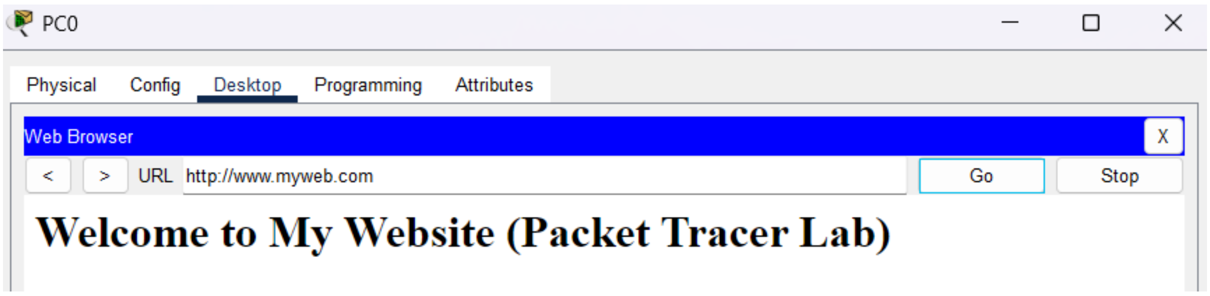
Name: www.myweb.com

Address: 192.168.1.3

Output:

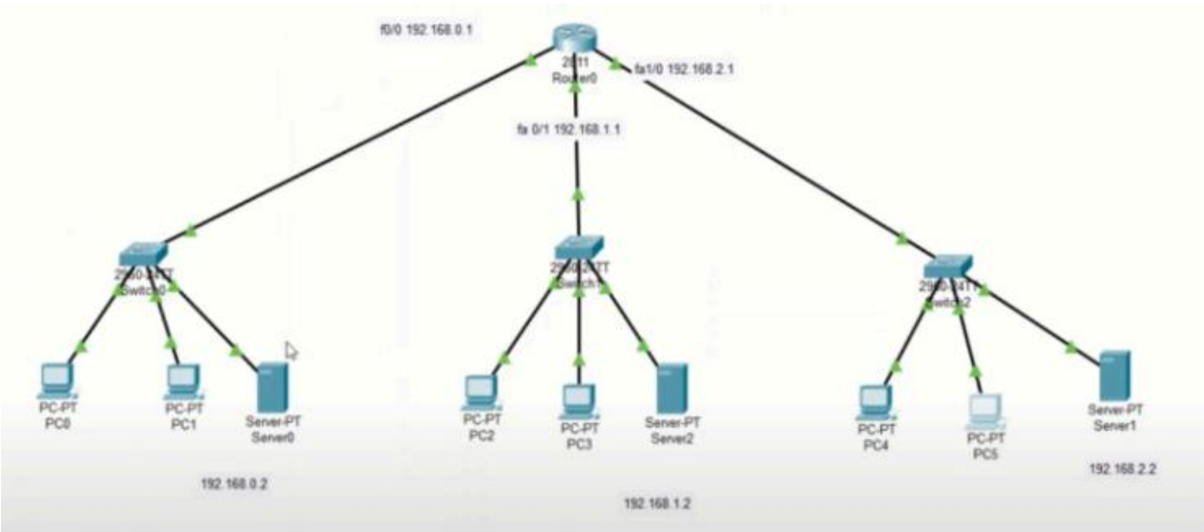

PC0
Physical
Config
Desktop
Programming
Attributes
Command Prompt
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.1.3
Pinging 192.168.1.3 with 32 bytes of data:
Reply from 192.168.1.3: bytes=32 time<1ms TTL=128
Reply from 192.168.1.3: bytes=32 time=1ms TTL=128
Reply from 192.168.1.3: bytes=32 time<1ms TTL=128
Reply from 192.168.1.3: bytes=32 time<1ms TTL=128
Ping statistics for 192.168.1.3:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 1ms, Average = 0ms
C:\>


Fire
Last Status
Source
Destination
Type
Color
Time(sec)
Periodic
Num
Edit
Delete
Successful
PC0
Server0
ICMP
0.000
N
0
(edit)


PC0
Physical
Config
Desktop
Programming
Attributes
Web Browser
X
<
>
URL
http://www.myweb.com
Go
Stop
Welcome to My Website (Packet Tracer Lab)

Q 2) i) Construct the below network in the Cisco packet tracer and configure DHCP, DNS, and HTTP protocols where DHCP, DNS, and Web server are on the different side of router

Diagram:



IP Config:

Router0:

FastEthernet0/0: 192.168.0.1 / 255.255.255.0

FastEthernet0/1: 192.168.1.1 / 255.255.255.0

FastEthernet1/0: 192.168.2.1 / 255.255.255.0

Server0: 192.168.0.2 / 192.168.0.1 / 192.168.0.2

Server1: 192.168.1.2 / 192.168.1.1 / 192.168.1.2

Server2: 192.168.2.2 / 192.168.2.1 / 192.168.2.2

Server0 DHCP Config:

Pool Name	Default Gateway	DNS Server	Start IP Address	Subnet Mask	Max User	TFTP Server	WLC Address
serverPool2	192.168.2.1	192.168.0.2	192.168.2.0	255.255.2...	256	0.0.0.0	0.0.0.0
serverPool1	192.168.1.1	192.168.0.2	192.168.1.0	255.255.2...	256	0.0.0.0	0.0.0.0
serverPool	192.168.0.1	192.168.0.2	192.168.0.0	255.255.2...	256	0.0.0.0	0.0.0.0

Server0 DNS Config:

No.	Name	Type	Detail
0	www.myweb.com	A Record	192.168.0.2

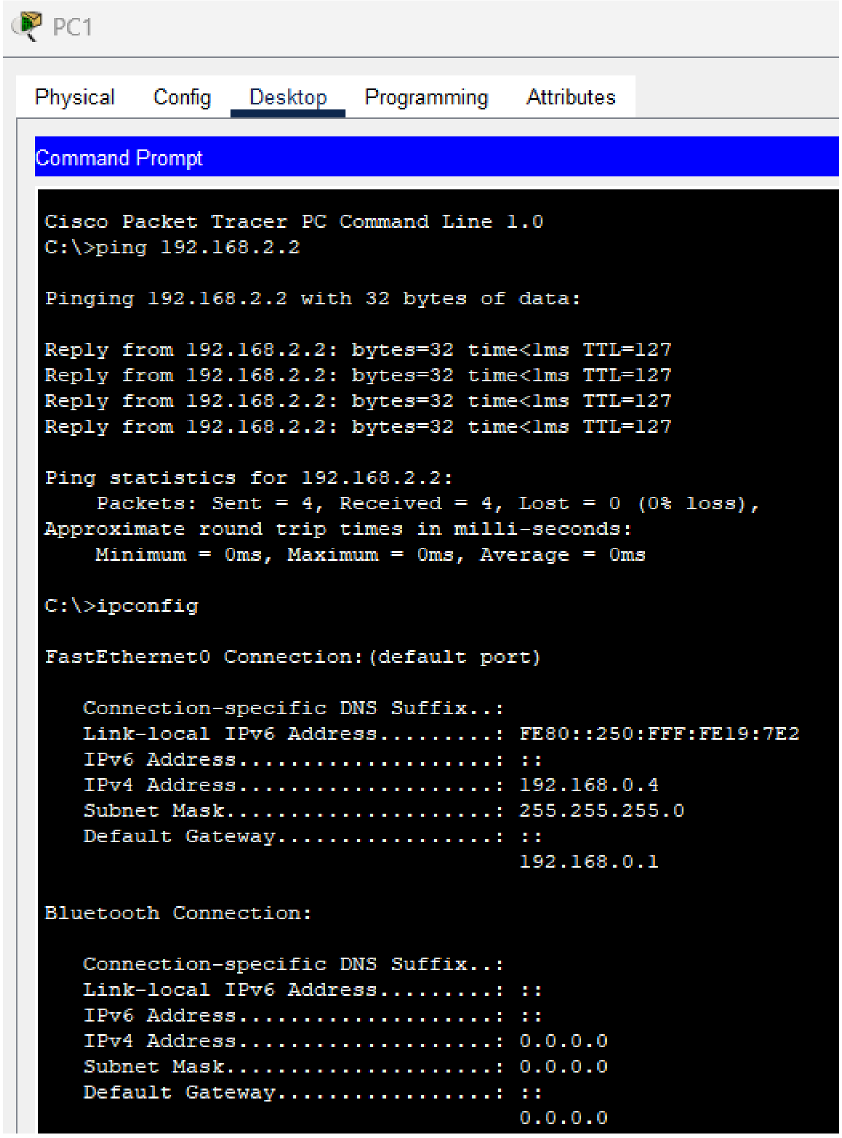
Router Config:

interface fa0/0
ip helper-address 192.168.0.2

interface fa0/1
ip helper-address 192.168.0.2

interface fa1/0
ip helper-address 192.168.0.2

Output:

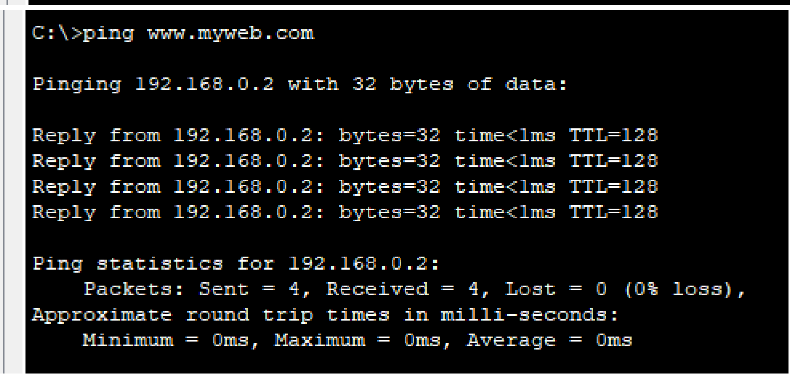


PC1

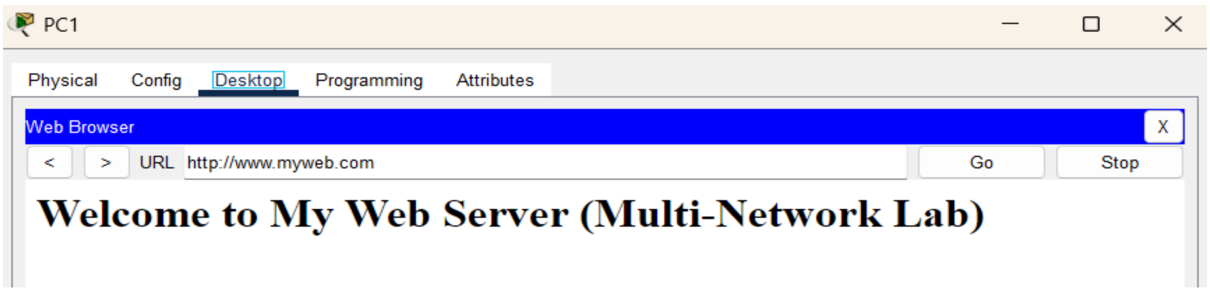
Physical | Config | Desktop | Programming | Attributes

Command Prompt

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.2.2

Pinging 192.168.2.2 with 32 bytes of data:

Reply from 192.168.2.2: bytes=32 time<1ms TTL=127
Reply from 192.168.2.2: bytes=32 time<1ms TTL=127
Reply from 192.168.2.2: bytes=32 time<1ms TTL=127
Reply from 192.168.2.2: bytes=32 time<1ms TTL=127

Ping statistics for 192.168.2.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ipconfig

FastEthernet0 Connection:(default port)

   Connection-specific DNS Suffix..:
   Link-local IPv6 Address.........: FE80::250:FFF:FE19:7E2
   IPv6 Address....................: ::
   IPv4 Address....................: 192.168.0.4
   Subnet Mask.....................: 255.255.255.0
   Default Gateway.................: ::
                                     192.168.0.1

Bluetooth Connection:

   Connection-specific DNS Suffix..:
   Link-local IPv6 Address.........: ::
   IPv6 Address....................: ::
   IPv4 Address....................: 0.0.0.0
   Subnet Mask.....................: 0.0.0.0
   Default Gateway.................: ::
                                     0.0.0.0
```



```
C:\>ping www.myweb.com

Pinging 192.168.0.2 with 32 bytes of data:

Reply from 192.168.0.2: bytes=32 time<1ms TTL=128
Reply from 192.168.0.2: bytes=32 time<1ms TTL=128
Reply from 192.168.0.2: bytes=32 time<1ms TTL=128
Reply from 192.168.0.2: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.0.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms
```



PC1

Physical | Config | Desktop | Programming | Attributes

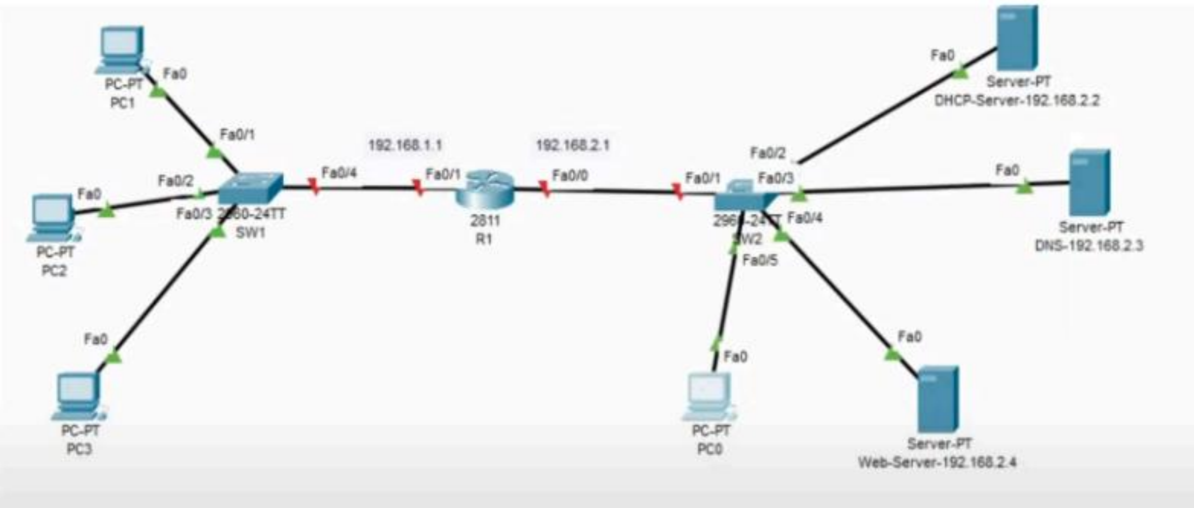
Web Browser

< > URL http://www.myweb.com Go Stop

Welcome to My Web Server (Multi-Network Lab)

ii) Construct the below network in the Cisco packet tracer and configure DHCP, DNS, and HTTP protocols where DHCP, DNS and Web server are on the same side of router

Diagram:



IP Config:

Router0:

FastEthernet0/0: 192.168.2.1 / 255.255.255.0

FastEthernet0/1: 192.168.1.1 / 255.255.255.0

Server0: 192.168.2.2 / 192.168.2.1 / 192.168.2.3

Server1: 192.168.2.3 / 192.168.2.1 / 192.168.2.3

Server2: 192.168.2.4 / 192.168.2.1 / 192.168.2.3

Server0 DHCP Config:

Pool Name	Default Gateway	DNS Server	Start IP Address	Subnet Mask	Max User	TFTP Server	WLC Address
LAN_POOL	192.168.1.1	192.168.2.3	192.168.1.10	255.255.2...	246	0.0.0.0	0.0.0.0
serverPool	0.0.0.0	0.0.0.0	192.168.2.0	255.255.2...	512	0.0.0.0	0.0.0.0

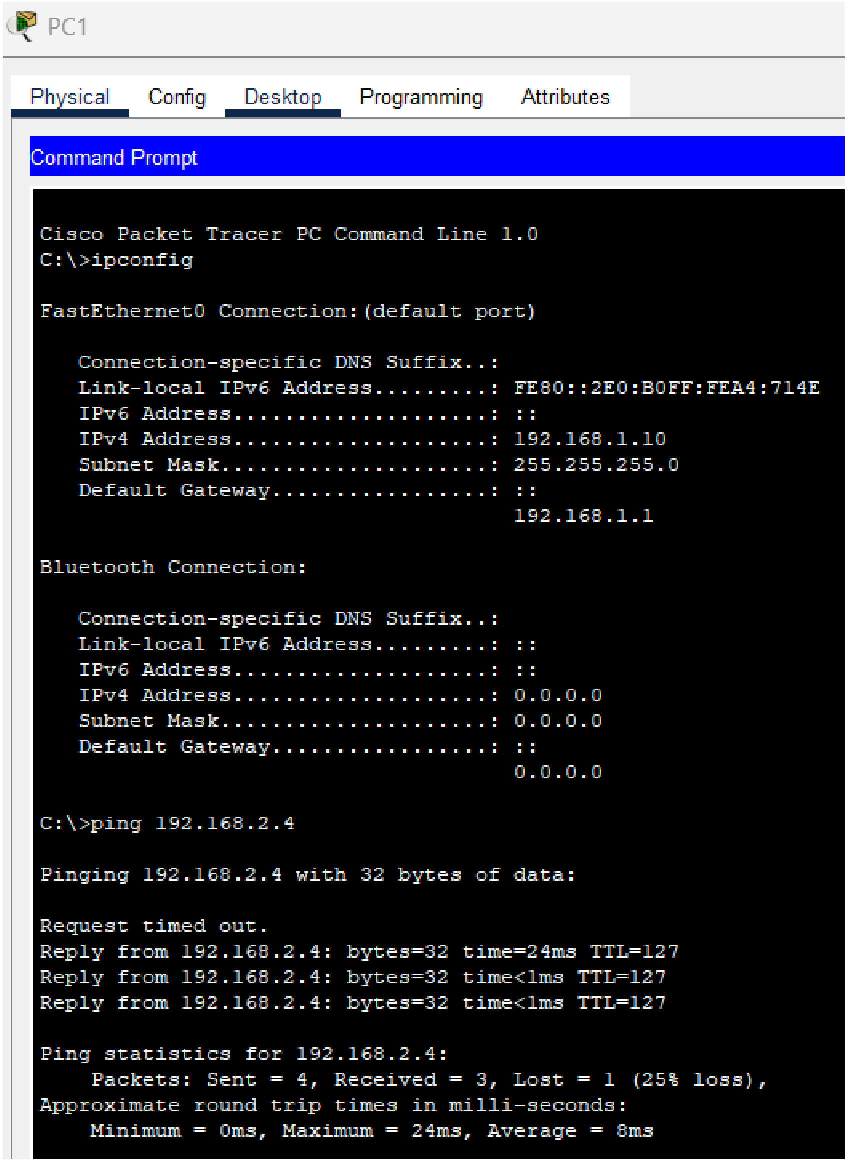
Server1 DNS Config:

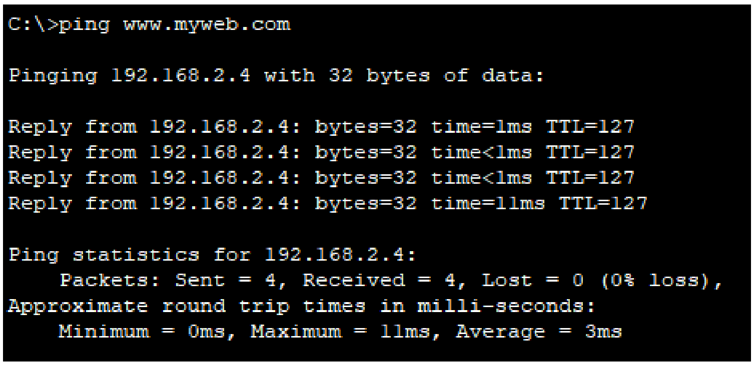
No.	Name	Type	Detail
0	www.myweb.com	A Record	192.168.2.4

Router Config:

```
interface fa0/0
ip helper-address 192.168.2.2

interface fa0/1
ip helper-address 192.168.2.2
```

Output:


PC1
Physical
Config
Desktop
Programming
Attributes
Command Prompt
Cisco Packet Tracer PC Command Line 1.0
C:\>ipconfig
FastEthernet0 Connection:(default port)
Connection-specific DNS Suffix..:
Link-local IPv6 Address.........: FE80::2E0:B0FF:FEA4:714E
IPv6 Address....................: ::
IPv4 Address....................: 192.168.1.10
Subnet Mask.....................: 255.255.255.0
Default Gateway.................: ::
192.168.1.1
Bluetooth Connection:
Connection-specific DNS Suffix..:
Link-local IPv6 Address.........: ::
IPv6 Address....................: ::
IPv4 Address....................: 0.0.0.0
Subnet Mask.....................: 0.0.0.0
Default Gateway.................: ::
0.0.0.0
C:\>ping 192.168.2.4
Pinging 192.168.2.4 with 32 bytes of data:
Request timed out.
Reply from 192.168.2.4: bytes=32 time=24ms TTL=127
Reply from 192.168.2.4: bytes=32 time<1ms TTL=127
Reply from 192.168.2.4: bytes=32 time<1ms TTL=127
Ping statistics for 192.168.2.4:
Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 24ms, Average = 8ms


C:\>ping www.myweb.com
Pinging 192.168.2.4 with 32 bytes of data:
Reply from 192.168.2.4: bytes=32 time=1ms TTL=127
Reply from 192.168.2.4: bytes=32 time<1ms TTL=127
Reply from 192.168.2.4: bytes=32 time<1ms TTL=127
Reply from 192.168.2.4: bytes=32 time=11ms TTL=127
Ping statistics for 192.168.2.4:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 11ms, Average = 3ms


PC1
Physical
Config
Desktop
Programming
Attributes
Web Browser
X
<
>
URL
http://www.myweb.com
Go
Stop
Welcome to Web Server (Same Network Lab)